FINE-TUNED UNDERSTANDING: ENHANCING SOCIAL BOT DETECTION WITH TRANSFORMER-BASED CLASSIFICATION

DEPARTMENT OF INFORMATION TECHNOLOGY

Guided By,
Dr.C. Aswini M.E.,Ph.D,

Team members:

Balaraman N - 71772218103
Logesh M - 71772218123
Ragul M - 71772218L05

ABSTRACT

- Malicious social bots impersonate humans, spread misinformation, and threaten digital discourse integrity on platforms like Twitter.
- Differentiating bot-generated text from human-written content is increasingly difficult due to advanced neural language models .
- We propose a transformer-based approach that fine-tunes Pre-trained Language Models on bot-specific tweet data, combined with a Feedforward Neural Network for binary classification.

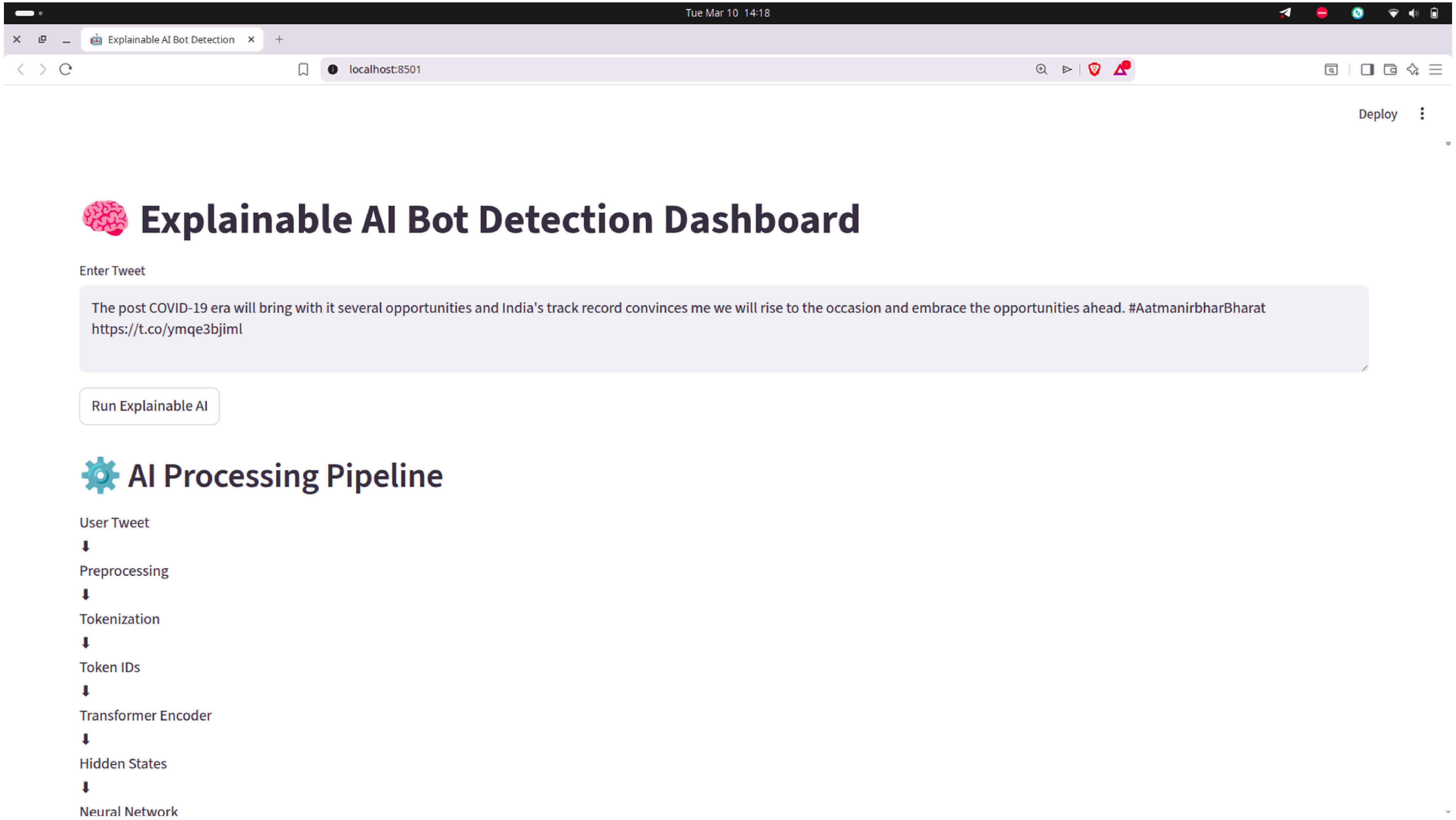
ACHIEVEMENT OF OBJECTIVES

- **Transformer-based Fine-Tuning** — Employed BERT, RoBERTa, DistilBERT, XLM-RoBERTa fine-tuned on bot-specific datasets (TweepFake & fox8-23) for contextual tweet embeddings
- **Tweet-Level Bot Classification** — Designed a Feedforward Neural Network (FNN) classifier achieving binary classification (Human vs. Bot) with high precision and recall
- **Explainability Integration** — Integrated SHAP, BERT Visualizer, and LLM-based prompting to interpret model decisions and enhance transparency

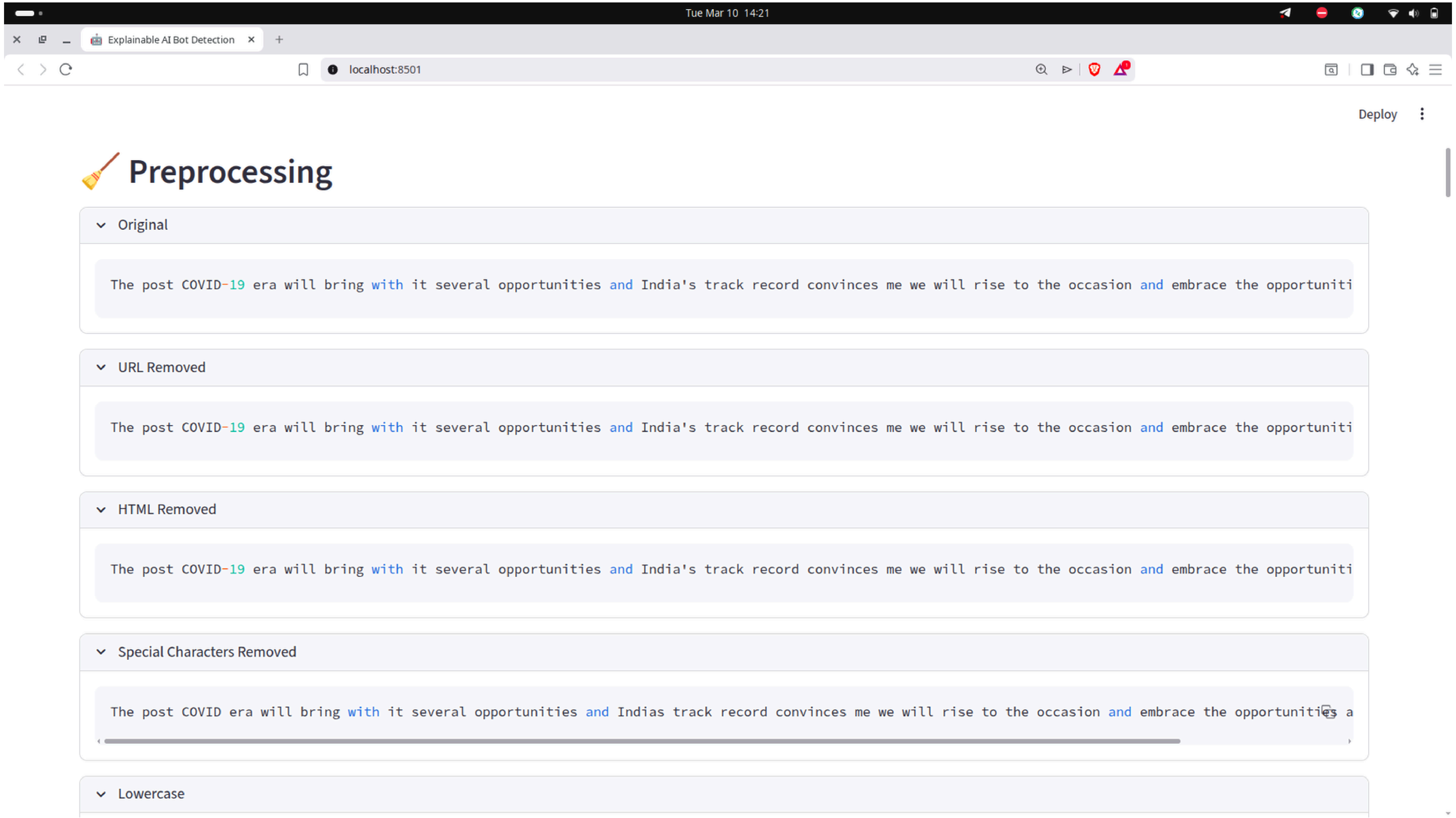
ACHIEVEMENT OF OBJECTIVES

Comparative Benchmarking	Systematically evaluated 6 PLMs against traditional embeddings (Word2Vec, GloVe + BiLSTM) across two diverse datasets
Domain Adaptation	Addressed the gap of bot-specific pre-training by fine-tuning on specialized bot-generated content datasets

DEMONSTRABLE RESULTS



DEMONSTRABLE RESULTS

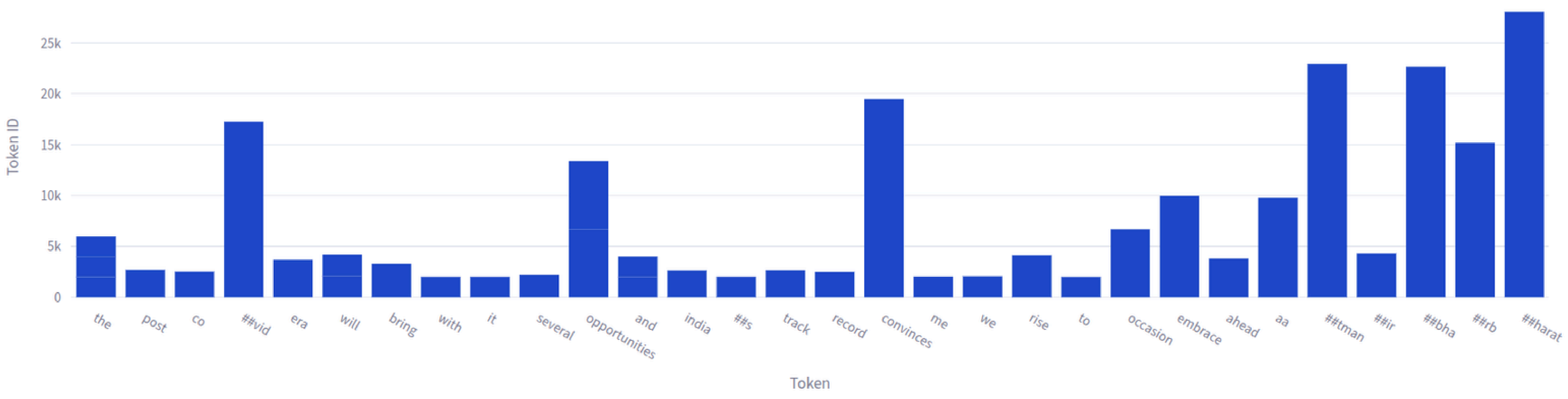


DEMONSTRABLE RESULTS

🔤 Tokenization

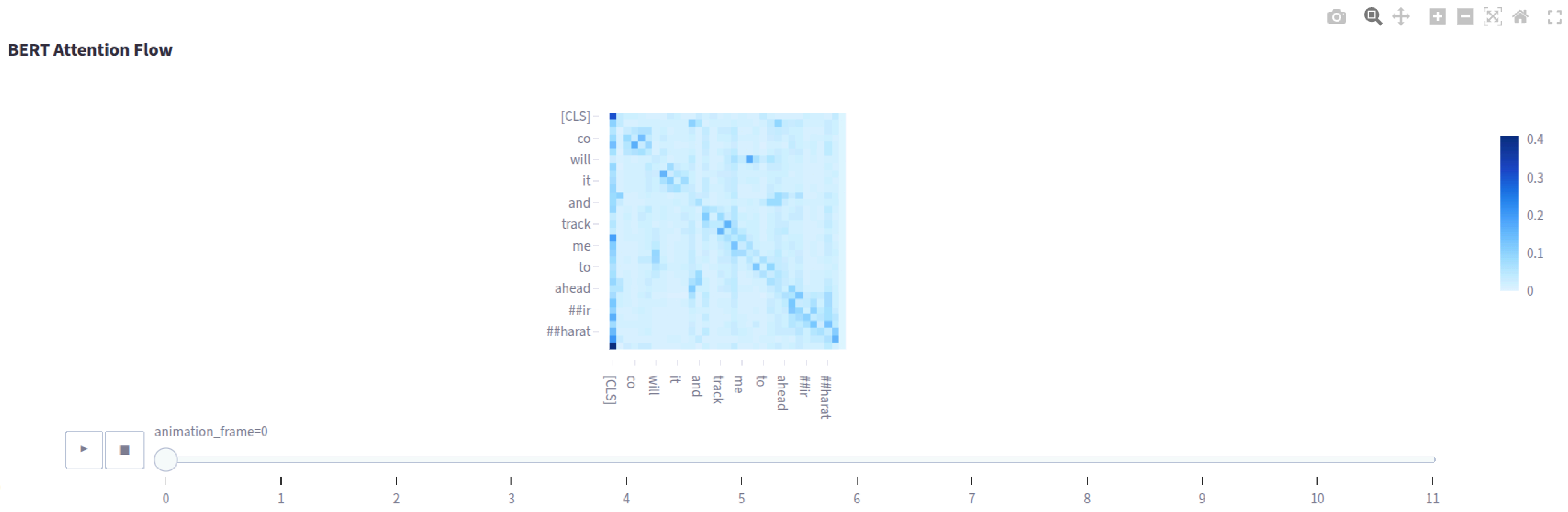
	Token	Token ID
0	the	1996
1	post	2695
2	co	2522
3	##vid	17258
4	era	3690
5	will	2097
6	bring	3288
7	with	2007
8	it	2009
9	several	2195

Token ID Distribution



DEMONSTRABLE RESULTS

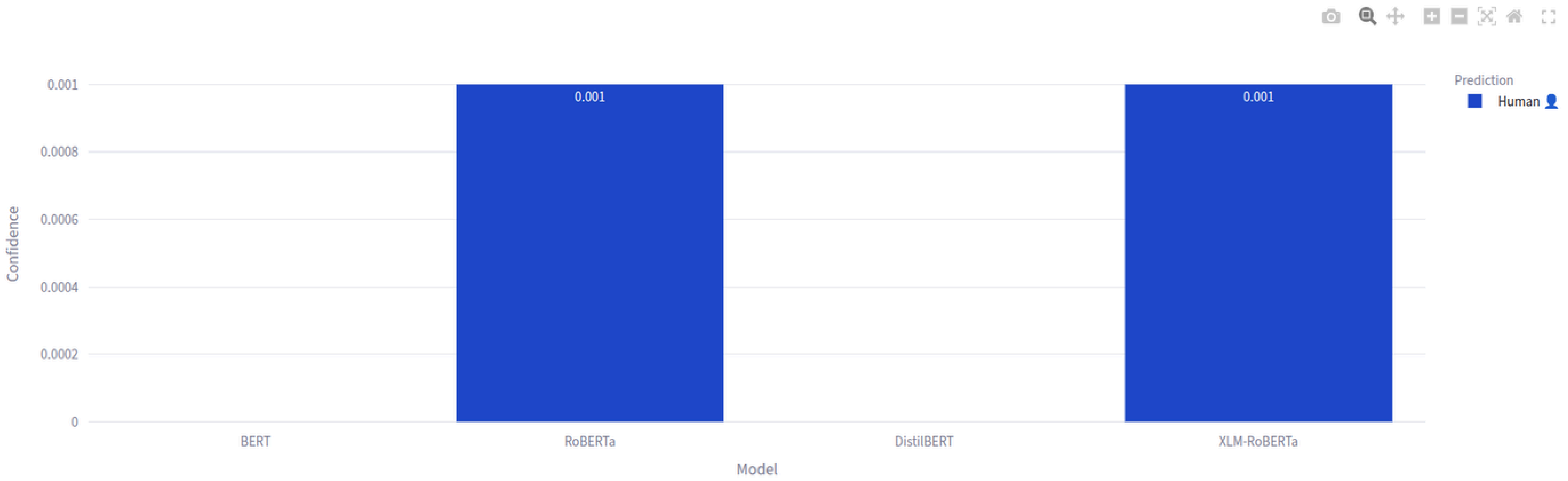
Attention Flow Across Layers



DEMONSTRABLE RESULTS

📊 Model Comparison

	Model	Prediction	Confidence
0	BERT	Human 👤	0
1	RoBERTa	Human 👤	0.001
2	DistilBERT	Human 👤	0
3	XLM-RoBERTa	Human 👤	0.001



🧠 Final Prediction

HUMAN ACCOUNT DETECTED 👤

INNOVATION & TECHNICAL CONTRIBUTION

PLM+FNN Architecture	First approach combining fine-tuned transformer embeddings with lightweight FNN for efficient tweet-level bot detection
Domain-Specific Fine-Tuning	Addressed literature gap: fine-tuned PLMs on bot-specific data instead of generic corpora, improving sensitivity to bot-like patterns
Multi-Dataset Validation	Validated approach on both synthetic (TweepFake) and real-world (fox8-23) datasets, ensuring generalizability
Comprehensive Comparative Study	Evaluated transformer models (BERT-base/large, RoBERTa, DistilBERT, XLM-RoBERTa) under identical hyperparameters for fair benchmarking

SOCIETAL IMPACT

Combating Misinformation	Reduces the spread of fake news, propaganda, and manipulated content by identifying bot-generated tweets at scale
Economic Integrity	Reduces fraud, fake engagement, and manipulative marketing - creating fairer digital economies for creators and businesses
Protecting Vulnerable Users	Shields younger audiences, elderly users, and marginalized communities from targeted bot campaigns, scams, and harmful content
Authentic Human Connection	Enables users to engage with real people, fostering genuine relationships and meaningful dialogue

FUTURE ENHANCEMENTS

- Multilingual PLMs for cross-language bot detection
- Combine content features with social graph data using TwiBot-20/22 datasets.
- Optimize models for real-time, low-latency deployment on edge/cloud platforms.
- Build adaptive online learning models to counter evolving AI-generated bots.

CHALLENGES

- **Computational Cost:** Use DistilBERT to reduce GPU usage.
- **Short Text Limit:** Optimize model for short tweets (≤280 chars).
- **Bot Evolution:** Use periodic retraining to detect new bots.
- **Domain Shift:** Support multiple platforms and languages.
- **Privacy & Ethics:** Ensure transparent and privacy-safe detection.

CONCLUSION

This project presents a transformer-based approach for detecting social bots on social media platforms. By fine-tuning pre-trained language models, the system effectively captures contextual differences between human and bot-generated content. The proposed method improves detection accuracy while reducing the need for manual feature engineering. Overall, the work contributes to building safer and more reliable social media environments.

Q/A

REFERENCE:

Fine-Tuned Understanding: Enhancing Social Bot Detection with Transformer-based Classification -

https://ieeexplore.ieee.org/document/10630818

THANK YOU!